

A Short Proof of 1-Y Well-Ordering in KP with ω_1

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Abstract

One elementary-substructure argument replaces the separate counterexample and extension operations of the longer proof. Only one recursively defined relation table is needed. The axiom bound remains Kripke–Platek set theory with infinity, full set induction, and an uncountable ordinal. The finite mountain reconstruction and copying results are cited from Phyrion’s supplied manuscript; the new ordinal argument is proved below. This is a mathematical proof using those results, not a new machine-checked proof.

1 The statement and the finite combinatorial input

Here KP_ω has Extensionality, Empty Set, Pairing, Union, Infinity, Δ_0 -Separation, Δ_0 -Collection, and full set induction. An *uncountable ordinal* means an ordinal $\nu > \omega$ for which no set function maps ω onto ν .

Use the inherited-ancestry 1-Y algorithm of [1, Sections 1–2]. Its legal expressions are

$$\mathcal{E} = \{\emptyset\} \cup \{(a_0, \dots, a_{m-1}) : 0 < m < \omega, a_0 = 1, 0 < a_i < \omega\}.$$

The operation E_N makes N additional copies of the bad part; E_0 deletes the last entry and $E_N(\emptyset) = \emptyset$. Put $t < s$ if $t \neq s$ and $t = E_N(s)$ for some $N < \omega$. Let $\text{Desc}(s)$ be the finite-step descendants of s , including s , and let $\mathcal{G}$ be the descendants of the seeds $(1, m)$, $m \geq 2$.

Theorem 1. *In KP_ω , the existence of an uncountable ordinal ν implies the existence of an ordinal $\chi \leq \nu$ and a set function $\mu : \mathcal{E} \rightarrow \chi$ such that*

$$\mu(\emptyset) = 0, \quad s \neq \emptyset \implies \mu(E_N(s)) < \mu(s) \quad (N < \omega). \quad (1)$$

Hence $<$ is well-founded, every expansion trajectory reaches $\emptyset$, and lexicographic order well-orders each $\text{Desc}(s)$ and $\mathcal{G}$.

Lexicographic order regards a proper prefix as smaller. Its well-ordering is asserted on the specified descendant sets, not on all of $\mathcal{E}$.

We state precisely the finite representation interface from [1, Sections 3, 6–7]. A rooted diagram has a width $m < \omega$ and finitely many atoms

$$(k, q, p, j), \quad k < \omega, \quad q \leq p < j < m.$$

Given an ordinal label set D and a relation $R(k, \eta, a, b)$, a representation of this diagram is a strictly increasing map $f : m \rightarrow D$ satisfying

$$R(k, f(q), f(p), f(j)) \quad ((k, q, p, j) \in A).$$

Write $\text{Rep}(A, f)$ for all these conditions. An endpoint template $\mathcal{N}$ is a finite set of triples (k, q, p) , $q \leq p < m$, and

$$\text{End}_b(\mathcal{N}, f) \iff \bigwedge_{(k, q, p) \in \mathcal{N}} R(k, f(q), f(p), b).$$

For $c < m$, its admissibility at (K, θ) is

$$\text{Adm}_{K, \theta}(\mathcal{N}, f, c) \iff \bigwedge_{(k, q, p) \in \mathcal{N}} [k < K \vee (k = K \wedge q < c \wedge f(q) < \theta)]. \quad (2)$$

Definition 2 (Reflection clause). The clause $\text{FR}_{K,\theta}(a, b)$ says: for every finite $A, \mathcal{N}$, every $c < m$, and every representation f of A with $f(c) = a$ and all $f(i) < b$, if $\text{Adm}_{K,\theta}(\mathcal{N}, f, c)$ and $\text{End}_b(\mathcal{N}, f)$, then there is a representation g of A such that

$$g \upharpoonright c = f \upharpoonright c, \quad g(i) < a \ (i < m), \quad \text{End}_a(\mathcal{N}, g). \quad (3)$$

The internal atoms of A may have arbitrary layer indices.

The complete finite mountain of s gives a diagram $A(s)$: an edge from column j to its parent p in layer k , with component root q , contributes (k, q, p, j) . The row number is not part of the atom. The mountain and each single expansion are finite: positive differences bound column heights, and extraction lowers every bottom value greater than 1 until an all-1 layer is reached [1, Lemma 2.1].

Lemma 3 (Finite copying interface). *Suppose $R(k, \eta, a, b)$ implies $\eta \leq a < b$, permits root weakening*

$$R(k, \eta, a, b), \ \xi \leq \eta \implies R(k, \xi, a, b),$$

and implies $\text{FR}_{k,\eta}(a, b)$ whenever $b \in D$. If $s \neq \emptyset$ has a representation with last label β , then each nonempty $E_N(s)$ has a representation with last label below β .

Proof. Use the finite copying invariant and canonical reconstruction of [1, Lemmas 2.2 and 7.1]. The original edges initialize the invariant, with the last label $\beta \in D$ as virtual endpoint. Deletion restricts the old representation. For a copy step, let f be the current width- n labeling, c its cut, and $\alpha = f(c)$; all labels are below β . Reflect the invariant's finite diagram, including retained source facts, to $g < \alpha$. Append the old tail:

$$h = g \frown (f(c), \dots, f(n-1)).$$

This is strictly increasing: g lies below α , and the appended tail starts at α . Its labels stay in D and below β . Old atoms hold by reflection, transferred atoms by retained source facts and root weakening, and seam atoms by the reflected endpoint template. The cited invariant updates the retained facts and templates for the next copy. Finite induction gives the copied diagram; canonical reconstruction identifies it with $A(E_N(s))$. $\square$

Thus the geometric case analysis remains an explicit cited input. We now construct the relation and initial representations needed to apply it.

2 One relation table

For now work in $\text{KP}_\omega + V = L$, with an uncountable ordinal. Let κ be the least one. There is a set function

$$e : \kappa \times \omega \rightarrow \kappa, \quad e(\alpha, \cdot) : \omega \twoheadrightarrow \alpha \quad (0 < \alpha < \kappa). \quad (4)$$

Indeed, Δ_0 -Collection gathers a set C containing a surjection for every positive $\alpha < \kappa$. A constructible level containing C supplies a canonical well-order of C . Bounded least choice in this well-order selects one surjection for each α , yielding e . Set $e(0, n) = 0$. This is the only use of $V = L$ in the construction before the final transfer.

Put

$$D = \{a : \omega < a \leq \kappa\}, \quad B = (\kappa + 1)^{<\omega}.$$

Define a single table R , up to upper endpoint κ , by

$$R(K, \theta, a, b) \iff [K < \omega, \ \theta \leq a < b \leq \kappa, \ \omega < a, \ \text{FR}_{K,\theta}(a, b)]. \quad (5)$$

The clause on the right uses the same R . Invalid tuples are false.

Lemma 4. *Equation (5) defines a set relation in KP. It has all the properties in Lemma 3.*

Proof. Recurse first on b , then on K , then on θ , defining all a -entries at one stage. Every query on the right of (5) is earlier. Internal input edges end below b ; all output edges end at or below $a < b$. An input endpoint edge ends at b , but (2) gives either a smaller layer or the same layer with a smaller root index.

To make this an actual ordinal recursion, set

$$L = (\kappa + 1) \cdot \omega, \quad \Gamma = L \cdot (\kappa + 1),$$

and place stage (b, K, θ) at

$$L \cdot b + (\kappa + 1) \cdot K + \theta < \Gamma \quad (\theta < b \leq \kappa).$$

Unused stages have empty rows. Diagram and template shapes are coded in ω ; label tuples and candidate outputs range over B . Given the earlier history, the next row is a boundedly defined subset of $\kappa + 1$, so exists by Δ_0 -Separation. Valid histories have a bounded verification predicate and agree on their common domains by induction. At limits, the derived Σ_1 -Collection scheme gathers the shorter histories; their compatible union is the next history. This constructs the full table in KP.

Strict endpoints and reflection follow from (5). If $\xi \leq \theta$, every template admitted at (K, ξ) is admitted at (K, θ) , so the latter reflection clause implies the former. This gives root weakening. $\square$

3 Initial representations from one elementary structure

Extend e to a total binary function on κ , with value 0 on unused inputs. Introduce the top-section predicate

$$P(K, \theta, a) \iff R(K, \theta, a, \kappa),$$

and consider the finite-language set structure

$$\mathcal{A} = (\kappa; <, R, P, e, \omega). \tag{6}$$

Here R is restricted to tuples from κ , and ω is a constant. The extra predicate P lets us discuss edges to κ while keeping the entire domain equal to κ .

Lemma 5 (Elementary initial segments). *For every $\gamma < \kappa$, there is δ with*

$$\max\{\omega, \gamma\} < \delta < \kappa, \quad \mathcal{A} \upharpoonright \delta \preceq \mathcal{A}.$$

Proof. KP constructs the satisfaction relation of the set structure $\mathcal{A}$: recurse over finite formulas and assignments, or use [2, Lemma 2.6] and the derived Δ_1 -Separation scheme. This concerns a set domain, not a truth predicate for the universe.

For each formula $\exists x \varphi(x, \vec{y})$, take its least ordinal witness in κ , or 0 if there is none. These countably many finite-arity Skolem operations have set graphs, using the satisfaction set and bounded separation. Close $\omega \cup \{\omega, \gamma\}$ under them and e . Call the resulting set H . Explicitly, H is the range of evaluation of finite closed terms in a countable language. Term codes lie in ω , and evaluation certificates lie in B ; bounded separation and finite induction therefore give a set surjection $\omega \twoheadrightarrow H$. The witness criterion for elementarity gives $\mathcal{A} \upharpoonright H \preceq \mathcal{A}$.

If $0 < \alpha \in H$, then (4) and closure give every $\xi < \alpha$ as $e(\alpha, n) \in H$. Thus H is an ordinal $\delta \leq \kappa$. It contains ω, γ , so both are below δ . Equality $\delta = \kappa$ would give a surjection $\omega \twoheadrightarrow \kappa$, a contradiction. $\square$

Call such a δ an *elementary height*. No collapse theorem, regularity theorem, or choice principle was used.

Lemma 6 (Two reflection observations). *For every elementary height δ ,*

$$P(K, \theta, a) \iff R(K, \theta, a, \delta) \quad (\theta \leq a < \delta, K < \omega), \quad (7)$$

$$P(K, \theta, \delta) \quad (\theta \leq \delta, K < \omega). \quad (8)$$

Proof. For (7), induct first on K , then on θ , simultaneously for all a with $\theta \leq a < \delta$. If $a \leq \omega$, both sides are false by the guard in (5). Assume $\omega < a$.

If $P(K, \theta, a)$ holds, take a demand at upper endpoint δ . Every input endpoint query is earlier in the induction by (2). The induction hypothesis changes these queries to P -queries, hence to edges ending at κ . We obtain a demand for $R(K, \theta, a, \kappa)$. Its output lies below a , and all its edge queries end at or below a ; it is therefore an output for the original demand at δ .

If $P(K, \theta, a)$ fails, validity and (5) give a finite counterexample to reflection at κ . Fix its finite diagram shape, template, and cut. For this shape, the assertion that a counterexample exists is a single first-order formula in $\mathcal{A}$: it quantifies the finitely many input labels, requires their internal R -edges and top P -edges, and denies the existence of a representing tuple satisfying (3). Its parameters are a, θ, ω and finitely many natural layer indices. All belong to δ . Elementarity supplies such a counterexample with all input labels below δ . The induction hypothesis changes its top premises to edges ending at δ . Candidate outputs still lie below a , and their queries end at or below a , so neither the candidates nor their truth conditions change. The same counterexample refutes $R(K, \theta, a, \delta)$. This proves (7).

To prove (8), take a demand with cut label $f(c) = \delta$ and upper endpoint κ . Fix only its prefix $u = f \upharpoonright c$; every coordinate of u is below δ . In $\mathcal{A}$, the original f witnesses the first-order assertion

$$\exists g_0 \cdots g_{m-1} \left[\text{Rep}(A, g) \wedge g \upharpoonright c = u \wedge \bigwedge_{(k,q,p) \in \mathcal{N}} P(k, g(q), g(p)) \right].$$

For the fixed finite shape this is a finite formula with parameters in δ . Elementarity gives a witness g entirely below δ . Equation (7) turns its top P -edges into edges ending at δ , so g satisfies (3). The formula does *not* fix $g(c) = \delta$ and does not use θ as a parameter; thus the case $\theta = \delta$ is included. $\square$

Corollary 7. *Every finite rooted diagram has a representation below κ .*

Proof. If $\alpha < \beta$ are elementary heights and $\theta \leq \alpha$, first apply (8) at α , then (7) at β :

$$R(K, \theta, \alpha, \beta).$$

For width m , finite iteration of Lemma 5 gives increasing elementary heights $\delta_0, \dots, \delta_{m-1}$. Set $f(i) = \delta_i$. Every atom (k, q, p, j) is satisfied by the displayed relation because $\delta_q \leq \delta_p < \delta_j$. Width 0 uses the empty representation. $\square$

4 The rank, well-ordering, and the final axiom reduction

Still in $\text{KP}_\omega + V = L$, define $\mu(\emptyset) = 0$ and, for nonempty s ,

$$\mu(s) = \min\{b < \kappa : (\exists f \in B) [\text{Rep}(A(s), f) \wedge f(|s| - 1) = b]\}. \quad (9)$$

Corollary 7 makes the set nonempty. Expression codes lie in ω ; their finite algorithm graphs are sets in KP. Thus (9) defines a set function by bounded separation on $\mathcal{E} \times \kappa$.

Choose a representation attaining $\mu(s)$. Lemmas 3 and 4 give every nonempty $E_N(s)$ a representation with smaller last label; all its labels remain below κ . Minimization gives

$$\mu(E_N(s)) < \mu(s).$$

For an empty output, the same inequality follows from $0 < \mu(s)$. This proves (1) under $V = L$.

Remove $V = L$. Ordinary KP has its constructible inner model $L \models \text{KP}_\omega + V = L$, with the same ordinals and natural numbers [3]. The hierarchy construction is available in KP [2, Theorem 2.8]. For its collection axiom, ambient Σ_1 -Collection gathers constructible witnesses together with levels containing them; one larger level bounds all witnesses. The other KP axioms relativize in the usual way, with full induction following from transitivity.

Now fix an ambient uncountable ordinal ν . It is also uncountable in L , since a surjection belonging to L would be a surjection externally. In L , perform the construction above at its least uncountable ordinal $\chi \leq \nu$, obtaining an actual set function $\mu : \mathcal{E} \rightarrow \chi$. The same μ satisfies (1) externally: finite expressions and their finite expansion computations are absolute, as are membership in this function and comparison of its ordinal values. We transfer the *rank witness*, not the assertion that L has no descending chain. Whether χ remains uncountable externally is irrelevant.

Well-foundedness and termination. For any nonempty ambient set $X \subseteq \mathcal{E}$, its set of μ -values has a least ordinal. An expression attaining it is $\prec$ -minimal by (1). If a set-coded trajectory never reaches $\emptyset$, its ranks strictly decrease; a least value among those ranks contradicts the next step. Neither argument requires dependent choice.

Lexicographic order. The finite numerical facts of [1, Lemma 9.1 and Section 9] are:

$$E_N(s) <_{\text{lex}} s \ (s \neq \emptyset), \quad i \leq j \implies E_i(s) \text{ is a prefix of } E_j(s),$$

every prefix is reachable by E_0 -steps, and

$$E_1(1, n+2) = (1, n+1) \quad (n < \omega).$$

Induct on $\mu(s)$ to compare two descendants of s by reachability. If one is s , comparison is immediate. Otherwise their paths begin at $E_i(s), E_j(s)$; assume $i \leq j$. The prefix facts make both expressions descendants of $E_j(s)$, whose rank is smaller, so induction applies. Consequently, on $\text{Desc}(s)$, strict lexicographic comparison is exactly nonempty nontrivial expansion reachability: the reverse direction would contradict lexicographic descent. Hence μ strictly preserves lexicographic order there.

Any two seeds have a common larger seed ancestor by the last displayed identity. Thus the same conclusion holds on $\mathcal{G}$. Minimizing ranks of arbitrary nonempty subsets now proves the asserted lexicographic well-orderings, completing Theorem 1.

What the simplification uses. The additional model-theoretic object is one ordinary satisfaction set for $\mathcal{A}$. KP supplies it, its least-witness operations, and their finite-term closure. The relation table still uses only bounded row definitions and collection of compatible histories. Thus the shorter proof keeps the same axiom bound; it establishes no claim of minimal proof-theoretic strength. The cited finite geometric proofs and this new set-theoretic proof have not been independently translated here into a checked KP or Lean derivation.

References

- [1] Phyrion. *Well-Ordering of the 1-Y Sequence System: Proof Exposition and Lean 4 Formalization*. Chinese manuscript supplied with this discussion, 10 September 2026. The finite results used here are in Sections 2–3, 6–7, and 9. <https://github.com/Phyrion1343/1Y-Well-Ordering-Lean>. Reported snapshot: `df3b69eea954c4640067e2b79d6c238a7217ce24`.
- [2] Zachiri McKenzie. *On the Relative Strengths of Fragments of Collection*. **arXiv:1806.08500v4**, 10 January 2019. <https://arxiv.org/pdf/1806.08500v4>. Lemma 2.6 (p. 6) gives Δ_1 -definability of satisfaction for set structures. **Theorem 2.8 (p. 7)** states that the constructible-hierarchy function is total and Δ_1 in KPI. These locators use the checked v4 PDF; its Lemma 2.7 is a different result about satisfaction in the theory **M**.
- [3] Michael Rathjen. *Power Kripke–Platek Set Theory and the Axiom of Choice*. 2018. <https://arxiv.org/abs/1801.01897>. Cited for the ordinary KP constructible-inner-model fact, not a Power-KP result.